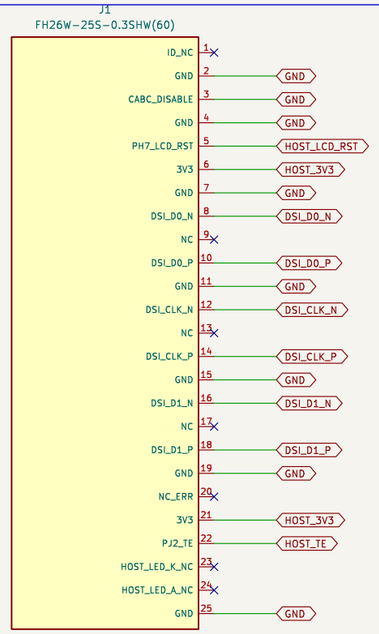
01 / STM32F469I-DISCO > DISEN DS070HSD39T-132

MIPI DSI and LCD interface | MB1189 B-07 / B-08 / C-01 pin assignments | Electrical revision I

01 / STM32 HOST - CN10



25-pin FPC / 0.30 mm pitch

HOST CONNECTION NOTES

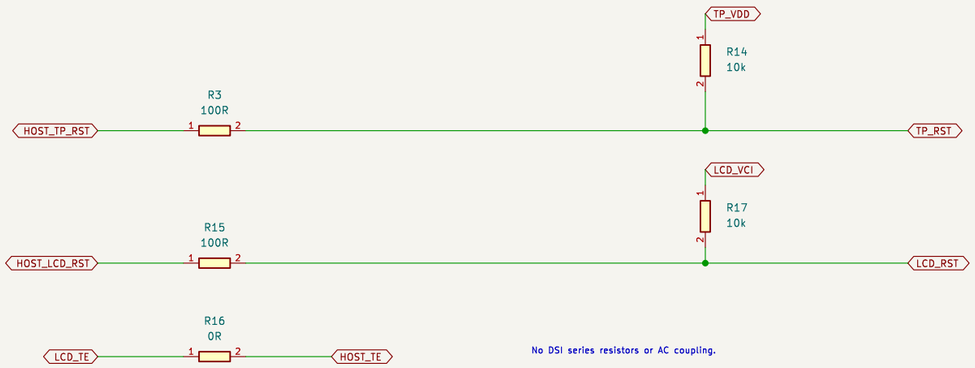
Remove original LCD and touch flexes.
Use a purpose-made 1:1 FPC; do not assume standard FFC pin order.

J1.23 / J1.24: host LED outputs unused.
J1.3 = GND holds onboard backlight EN low via R117. Confirm R117=0R and R119=DNP; do not drive PA3 high.

02 / MIPI DSI - DIRECT PAIRS

NET / PAIR	HOST J1	LCD J2
DSI_D0_P	10	16
DSI_D0_N	8	17
DSI_D1_P	18	20
DSI_D1_N	16	21
DSI_CLK_P	14	24
DSI_CLK_N	12	25

03 / RESET & FRAME SYNCHRONIZATION



No DSI series resistors or AC coupling.

05 / POWER, TOUCH & BACKLIGHT SHEETS

Logic power & touch

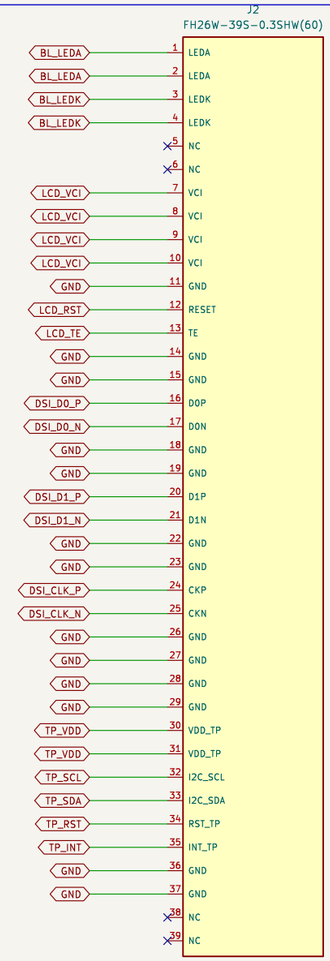
File: Logic_Touch.kicad_sch

Backlight / external power

File: Backlight.kicad_sch

Logic: host 3.3 V only. Boost: separate regulated 5 V.
All blocks share GND. Identical net labels are connected.

04 / DISEN LCD - 39-PIN FPC



0.30 mm pitch / verify contact side & pin 1

MIPI D-PHY: three direct pairs, continuous GND.
PCB + flex differential impedance: 100 ohm.
No DSI stubs/testpoints; no generic TVS on DSI.
Check FPC drawings and end-to-end continuity.

ENGINEERING PROTOTYPE / Not approved for manufacturing.
Layout edition changes presentation only; component values, DNP states and connectivity remain revision I.

Presentation update only | Electrical revision I retained

Yakup Turkan

Sheet: /
File: STM32F469_DISEN_RevI.kicad_sch

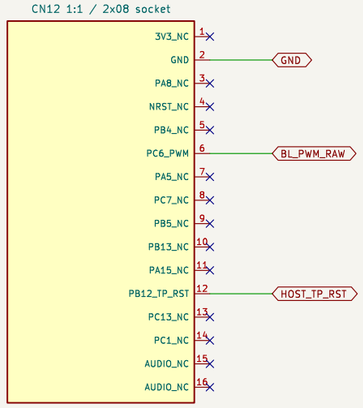
Title: STM32F469 / DISEN - MIPI & LCD interface

Size: A3	Date: 2026-10-16	**Rev: I / Layout 1**
KiCad E.D.A. 9.0.2		Id: 1/3

02 / LOGIC POWER, TOUCH & HOST CONTROL

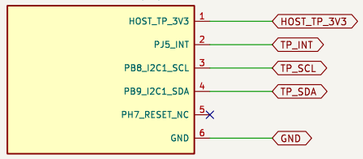
3.3 V logic domains | LCD and touch reset kept independent | Test points grouped by function

01 / STM32 CONTROL – CN12



2x8 socket / 2.54 mm / 1:1 mapping
Follow physical pin 1 for socket orientation.

02 / TOUCH HOST – CN2



6-pin FPC / 0.50 mm pitch
J3.1 supplies TP_VDD through R5.
J3.5 unused: LCD and touch resets independent.

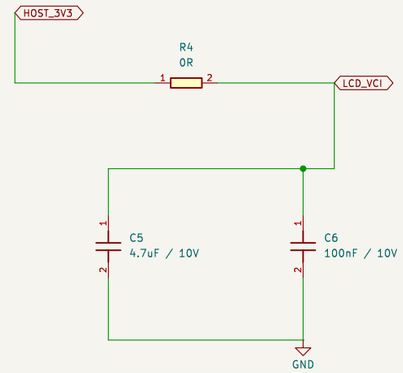
INTERFACE CONVENTIONS

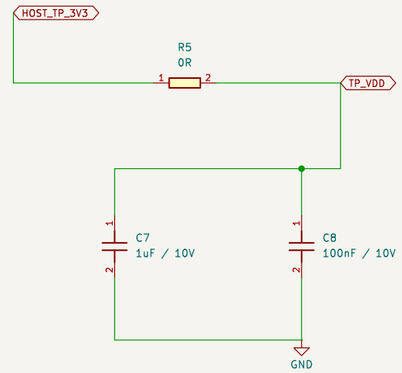
Host LCD 3.3 V and host touch 3.3 V
remain separate at the input connectors.

BL_PWM_RAW: PC6 from CN12.
HOST_TP_RST: PB12 from CN12.
TP_INT: PJ5 from CN2.

All signal names match electrical revision I.

03 / LOGIC SUPPLIES & DECOUPLING





LCD_VCI / host LCD 3.3 V

TP_VDD / host touch 3.3 V

Keep 5 V away from both logic rails. Current budget and voltage tolerance require validation.

04 / TOUCH I2C – OPTIONAL PULL-UPS



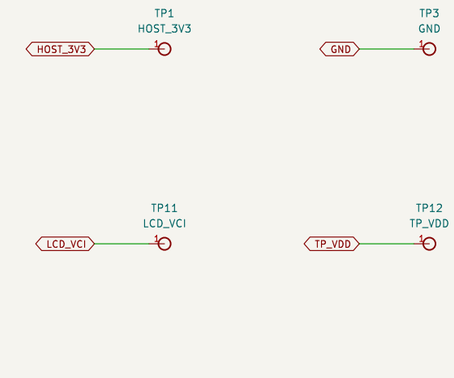


PB8/PB9 = I2C1, open-drain. PJ5 = touch interrupt.
Start at 100 kHz; verify ILI2130 address and pull-up load.
No additional INT pull-up: MB1189 R36 is 4.7k to 3V3. R1/R2 remain DNP.

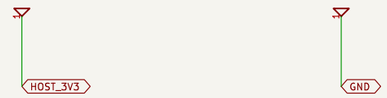
VALIDATION NOTES – REVISION I

Power budget is not proven: LCD and touch current = TBD
in the supplied preliminary datasheet. Measure before release.
Touch 2.8–3.3 V range / 3.3 V nominal: confirm tolerance.
Do not connect 5 V to LCD_VCI or TP_VDD.

05 / LOGIC TEST POINTS



Supply declarations



Presentation update only | Electrical revision I retained

Yakup Turkan

Sheet: /Logic power & touch/
File: Logic_Touch.kicad_sch

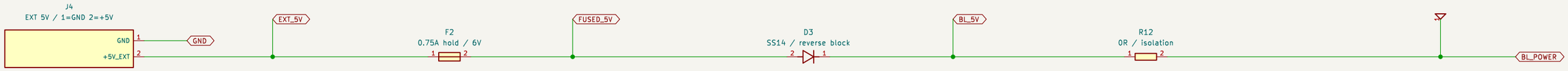
Title: STM32F469 / DISEN – Logic power & touch

Size: A3	Date: 2026-10-16	**Rev: I / Layout 1**
KiCad E.D.A. 9.0.2		Id: 2/3

03 / EXTERNAL 5 V & TPS61165 BACKLIGHT

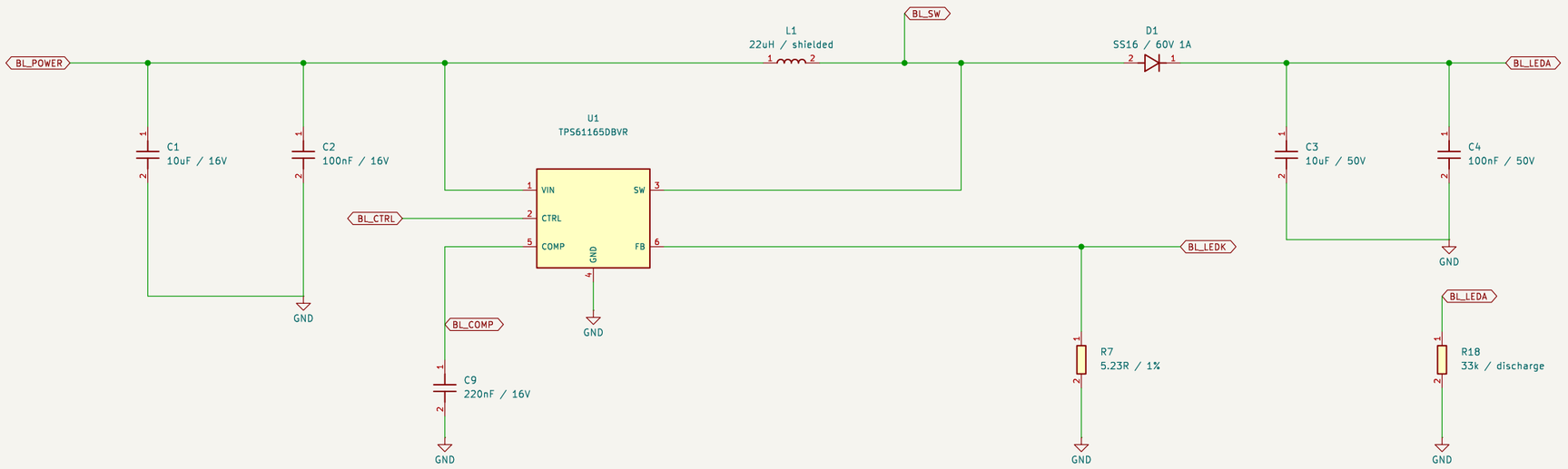
LED load: 21.6–27.2 V at 40 mA | Separate regulated boost supply | Common GND with F469

01 / INPUT POWER – FUSE, REVERSE BLOCK & ISOLATION



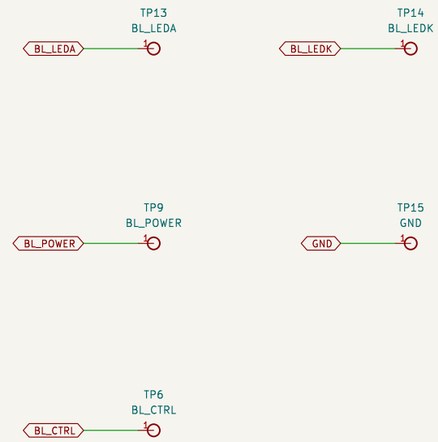
External regulated 5 V is mandatory. J4.1=GND / J4.2=+5V. No connection to host +5 V or host LEDA/LEDK.

02 / BOOST POWER STAGE & LED CURRENT RETURN

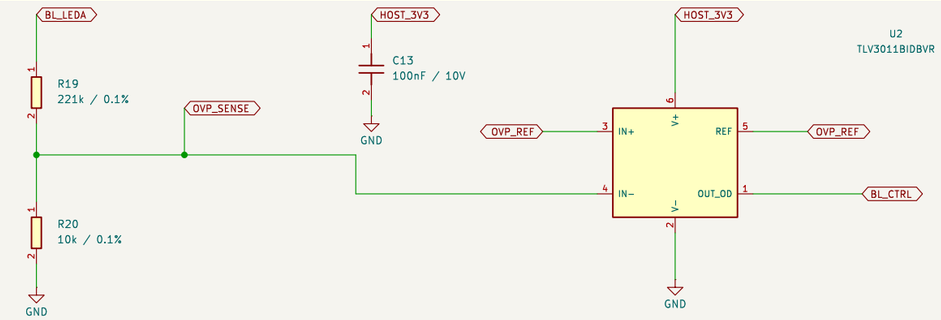


I_LED = 0.200 V / 5.23 ohm = 38.24 mA nominal; target <40 mA. LEDK connects to U1.FB and R7 only.

03 / BACKLIGHT TEST POINTS

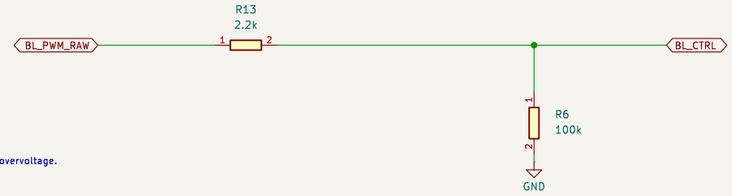


04 / OVP SENSE & COMPARATOR SHUTDOWN



REV I VALIDATION: OVP nominal = 1.242 x (1 + 221k/10k) = 28.690 V. HOST_3V3 must settle >=10 ms before PWM.
Open-load can hiccup; measure peak <30 V. CTRL shutdown is not immediate (2.5 ms); <30 V OVP is NOT certified.
Do not hot-plug. Turn PWM off before panel disconnect. Verify SW loop, GND return, EMI and thermal.
Fuse is not an overvoltage limiter. Component values, LED current and protection require prototype validation.

05 / PWM CONTROL & DEFAULT-OFF



U2 pulls CTRL low during overvoltage.

Presentation update only | Electrical revision I retained

Yakup Turkan

Sheet: /Backlight / external power/
File: Backlight.kicad_sch

Title: STM32F469 / DISEN – External 5 V & backlight

Size: A3	Date: 2026-10-16	**Rev: I / Layout 1**
KiCad E.D.A. 9.0.2		Id: 3/3